

In vitro anti-metastatic activity of enterolactone, a mammalian lignan derived from flax lignan, and down-regulation of matrix metalloproteinases i...

BACKGROUND: Actin cytoskeleton is involved in actin-based cell adhesion, cell motility, and matrix metalloproteinases (MMPs). MMP2, MMP9, MMP11 and MMP14 are responsible for cell invasion in breast cancer metastasis. The dietary intake of lignan from flax seed gets converted to enterolactone (EL) and enterodiol in the human system. Here we show that the enterolactone has a very significant anti-metastatic activity as demonstrated by its ability to inhibit adhesion and invasion and migration in MCF-7 and MDA MB231 cell lines. **MATERIALS AND METHODS:** Migration inhibition assay, actin-based cell motility assay along with reverse transcriptase polymerase chain reaction (RT-PCR) for MMP2, MMP9, MMP11 and MMP14 genes were performed in MCF-7 and MDA MB 231 cell lines. **RESULTS:** Enterolactone seems to inhibit actin-based cell motility as evidenced by confocal imaging and photo documentation of cell migration assay. The results are supported by the observation that the enterolactone in vitro significantly down-regulates the metastasis-related metalloproteinases MMP2, MMP9 and MMP14 gene expressions. No significant alteration in the MMP11 gene expression was found. **CONCLUSIONS:** Therefore we suggest that the anti-metastatic activity of EL is attributed to its ability to inhibit cell adhesion, cell invasion and cell motility. EL affects normal filopodia and lamellipodia structures, polymerization of actin filaments at their leading edges and thereby inhibits actin-based cell adhesion and cell motility. The process involves multiple force-generating mechanisms of actin filaments i.e. protrusion, traction, deadhesion and tail-retraction. By down-regulating the metastasis-related MMP2, MMP9 and MMP14 gene expressions, EL may be responsible for cell invasion step of metastasis.